

Machine Learning Final Assignment: A Study of NFL Play-by-Play Data from 2009 – 2018 and Predicting Expected Points Added Per Play

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CMPS 5323: Machine Learning
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December 10, 2025

1. Introduction

As a sports fan, it's my pleasure to perform a project on one of my favorite games. I grew up watching football with my dad. The logos and uniforms drew me in, but the game always fascinated me. The goal of trying to maintain possession of a ball and keeping it moving down the field towards an end zone that was defended by the other team was just simple enough to understand but complicated enough to make it interesting. I remember asking my dad question after question to get an understanding of the rules and eventually answering those same questions for my own sons as they grew up watching with me.

As time has passed and evolved, so has football. Tactics within the rules change year-by-year, and those tactics that make the game less enjoyable for the fans drive changes to the rules. Specifically, fans tend to enjoy games where points are scored, especially American fans. Football is one of the sports that has seen some of the biggest changes to help the offense score points of any sport in the American landscape. Football has also been the beneficiary of recent advancements in machine learning and artificial intelligence in many aspects, from predicting points scored by a team to movement by an individual player.

There are so many interesting paths that this project can take, but ultimately, I am restricted to a semester to perform research, experiment, and report on my findings. Player type prediction was one path that I considered, however the data manipulation that would be required seems to be above my current skills and would require more time than I have to solve. Additionally, player evaluations may require more player data than I'll be able to obtain, as the NFL typically keeps that information for their Big Data Bowl events and only releases it to participants.

As such, my objectives in this project are to create a model that accurately predicts expected points added on a per-play basis. I intend to use only information that is available before the play begins to predict the change in expected points after the play.

My intent with this research is to gain a greater understanding of machine learning and artificial intelligence. So much recent research has gone into this area by so many people that have a greater understanding of the subject, that I don't think I'll be able to make any real advancements, but at least I can advance my own understanding.

2. Background

Parth Patel [1] is one of those that did similar research using the Max Horowitz dataset, but he set out to predict play calls between run and pass

based on information such as down, distance, score, and team tendencies. Using XGBoost, he was able to predict correctly between run and pass with 73% accuracy. Additionally, Avery Ennis used a gradient boosted decision tree and a neural network to predict yards gained and expected points added per play [2]. Using Mean Average Error, his models were able to obtain an error of 6.1 yards and 1.05 expected points per play. He was disappointed with the results, and this makes sense. Anecdotally, to highlight the significance of the error, I learned at a young age that a good average for a running back is approximately 4 yards per carry. This is because it gets the team in possession of the ball almost halfway to 10 yards, which resets the number of chances they have to go 10 yards further. Naturally, three runs of that distance keep the possessing team ahead of the pace to gain 10 yards within three plays and keeps their drive down the field progressing. An error larger than the value of yards that a team hopes to gain on any play is highly inaccurate for predicting yards gained per play. After all, football is often referred to as a game of inches.

However, this error highlights the difficulty of predicting what might happen in the next play. Davis, et al. listed multiple reasons for that fact. For one thing, we as fans and analysts can't really know where a player is supposed to be or what they are supposed to do. For another, things change quickly year-to-year and even game-to-game in sports. Finally, every game consists of a small, limited number of plays, and players receive an even more limited time in the spotlight within the frame of the game [2].

Blake, Abud, David, and Pasteur sought a similar goal to mine and attained a mean absolute error of 11.24 points per game for the four seasons between 2007 and 2010 [5]. They tested three models with different sets of statistics that used a

4-neuron hidden layer with the Tan-Sigmoid activation function followed by a 6-neuron hidden layer utilizing the same function. This layer was fed into a mean squared function for backpropagation to train the model. A fourth model used a technique that they referred to as linear regression combinatorial optimization that they fed all sets of statistics into. Finally, a fifth model used a similar fully connected neural network, but with 4 neurons in the first hidden layer and 8 neurons in the second hidden layer and was fed with all sets of statistics. Between the five models, two were able to achieve a mean absolute error of 11.24 points per game, and a third was right behind them at 11.25. However the worst performance came from the fifth model, and it achieved 11.41, so the error between the best and worst performing models was merely 0.17 points. This is the approach that I intend to use for my research.

With so many prediction metrics out there, I wanted to take a minute to explain expected points added versus win probability added. The difference is that they ask different questions [6]. The expected points model asks, "how many points is the current game situation worth, in expectation, conditional on the features of that game situation (e.g. down, distance, yard line, score differential, time remaining, etc)." The win percentage model asks, "how likely is it that the possession team will win the game, conditional on the features of that game situation." With these things in mind, I will be sure to include that information in creating my model.

3. Methodology

The dataset that I'll be using has been used in a few instances before, as mentioned previously. It was originally published by Max Horowitz in 2016 and updated twice afterwards. It contains

100 features of NFL play-by-play data from 2009 until 2018 [4].

My first step with the dataset was to drop out all the columns that were either duplicated or contained information about the play that was about to happen. Next, all values in the dataset that were not a number were either removed, in the case of the down, expected points added, and play time differential columns, or converted from a string value to a numerical value, like in the case of timeouts, home and away teams, and possession and defensive teams. While it removed many of the unnecessary features, this process created two new binary features for timeouts by the offense or defense, two new binary features detailing whether the home team was on offense or defense, and 64 new Boolean features detailing whether each of the 32 teams in the league were the home or away team in the game. This resulted in a slight decrease from the original 101 features to 95. Finally, I separated the dataset by game, so that it would be fed into each model in a chronological order and normalized the feature values using Z-score normalization.

My initial thought was that I wanted to train some models using a selection of games with all the data from the data set that could impact expected points added. I would take out anything from the play that contributes to the calculation of expected points in the prediction phase. It was important that the data be presented to the model in order, as that is how the game is played, with late game plays are impacted by early game plays. The contestants for each game were important information, because, especially as seasons progress, a team's performance should become more predictable. According to [3], which focuses on the consistency of a player's skills, performance is typically consistent over the course of a season or two. Extrapolating this

across the entire team, I intend to restrict the data used to train one model to two years.

Ultimately, I generated five different models for comparison. All models used a split of 70% training data, 15% validation data, and 15% testing data. The first three models used the entire dataset; the fourth model used the first two years of game identifiers from the dataset; and the fifth model used all game identifiers that included one team as the home or away team. This was to see if removing the noise of all the other teams' data helped improve model accuracy. The team that I chose to use for the model was Washington, and it was for no other reason than that they are my favorite team.

Model construction was as follows, and is detailed in table 2, section 8 of this report. All models contained an output layer utilizing a single, fully connected neuron, linear activation, AdaM optimization, and Mean Squared Error for loss.

Model #1 used a smaller construction like that described in [5], although it was more in-line with modern neural networks than the one illustrated there, containing 16 fully connected neurons in the first hidden layer and 8 fully connected neurons in the second hidden layer. Model #2 contained 3 ReLu-activated hidden layers of 128, 64, and 32 neurons. Like model #1, model #3 contained 2 ReLu-activated hidden layers, but the number of neurons was reduced to four and six. Models #4 and #5 were modeled as closely to [5] as I could manage in Google Colab, with 2 TanH-activated hidden layers of four and 6 neurons.

4. Results

Final model results are quantified in table 3, section 8 of this report using mean absolute error to quantify the error in number of points per play

and game. The results indicated within are the values of the test split. Of the five models, model 4 performed best on per play average, and model 3 performed best on per game average. However, model #3 was closer to model #4 measured per play than model #4 was to model #3 measured per game. Additionally, the standard deviation for model #3 was lower on both counts.

The difference between the best per play result and the worst per play result was 0.05490. Removing model #5, which performed drastically worse than the other four, that difference drops to 0.005100. By per game average, that difference is 0.98.

Test split loss is quantified in the following table using mean squared error. Loss for the first 3 models was just under 1.96, for the most part. Model #4 runs came out just over 1.88, and model #5 runs came out within approximately 0.0201 points of 2.2160.

	Test Loss		
Model #	Run #1	Run #2	Run #3
1	1.9580	1.9584	1.9591
2	1.9613	1.9511	1.9561
3	1.9569	1.9598	1.9631
4	1.8848	1.8814	1.8800
5	2.1971	2.2160	2.2361

Table 1: Model Test Loss

In the interest of space, training and validation loss images included within this report will be limited to the first runs of all models, as each run was similar, and will be included in the submission package with the .ipynb files. See figures 1 through 5, below, for those images.

Attempting to find the features that the models found to be most important was something that I felt it important to ascertain, as well. Figures 6 through 10 illustrate the results from the first run of each model. Although there were differences

from the first run to the next, the results are similar enough to only highlight the difference between each model for the purposes of this report.

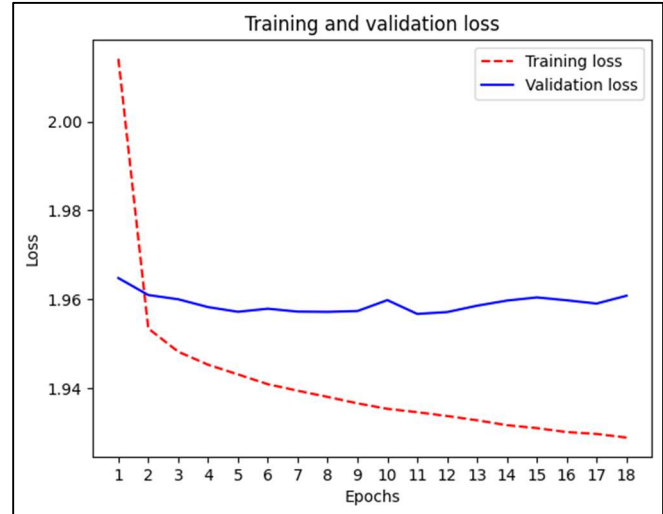


Figure 1: Model #1 Training vs. Validation Loss

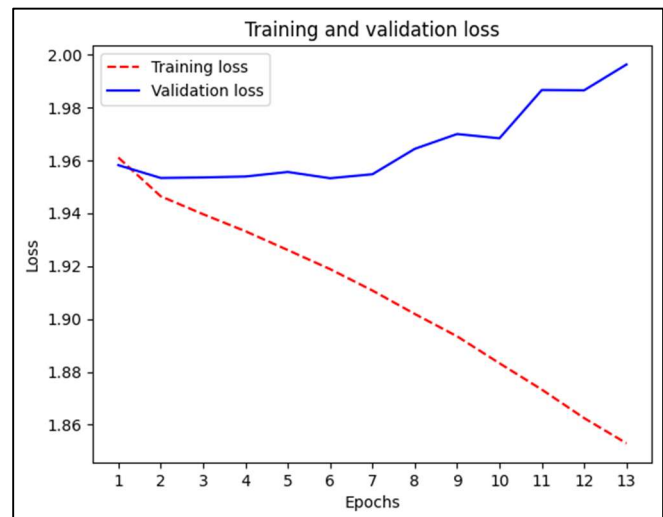


Figure 2: Model #2 Training vs. Validation Loss

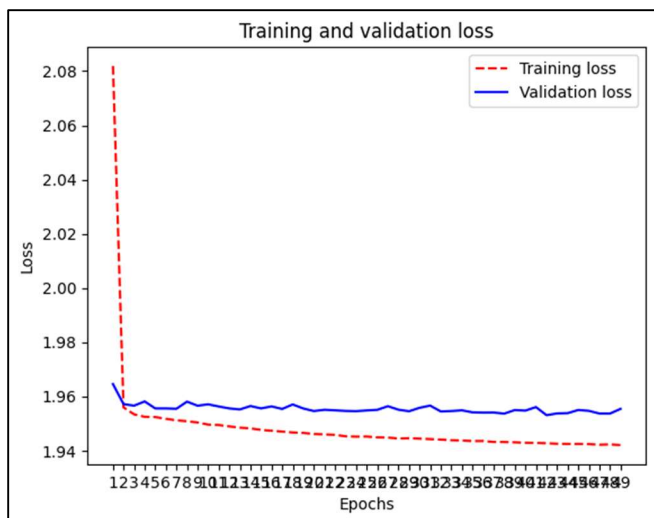


Figure 3: Model #3 Training vs. Validation Loss

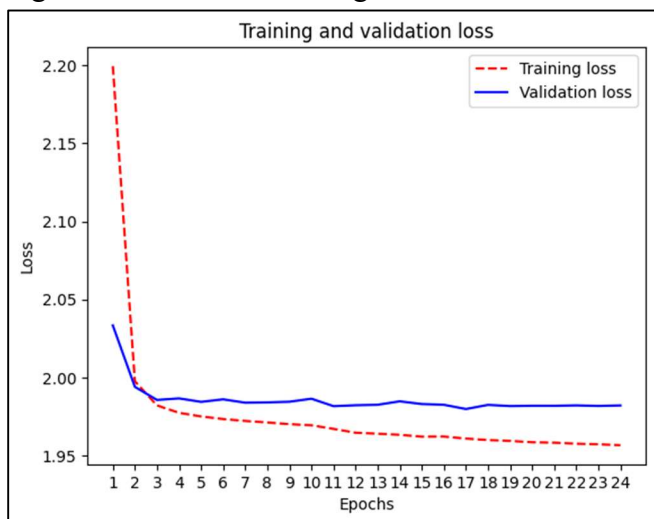


Figure 4: Model #4 Training vs. Validation Loss

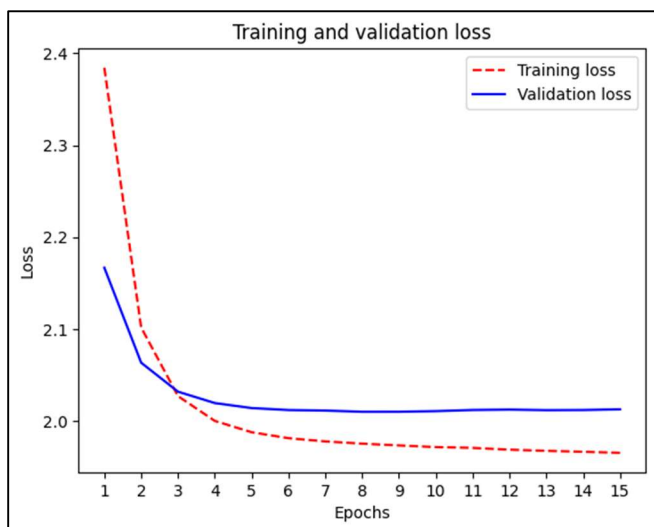


Figure 5: Model #5 Training vs. Validation Loss

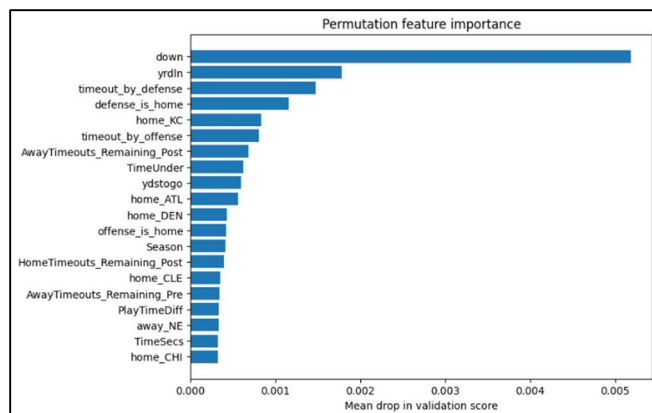


Figure 6: Model #1 Feature Importance

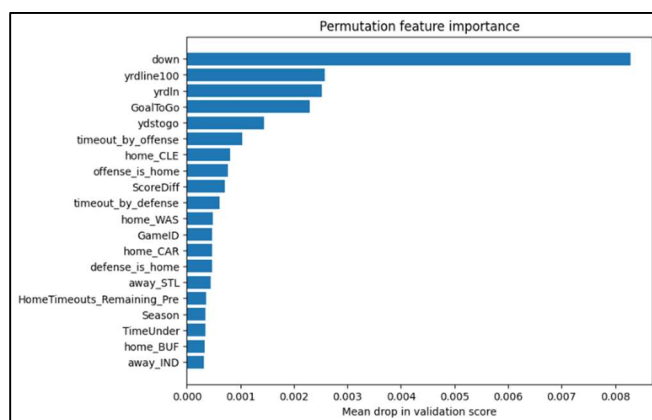


Figure 7: Model #2 Feature Importance

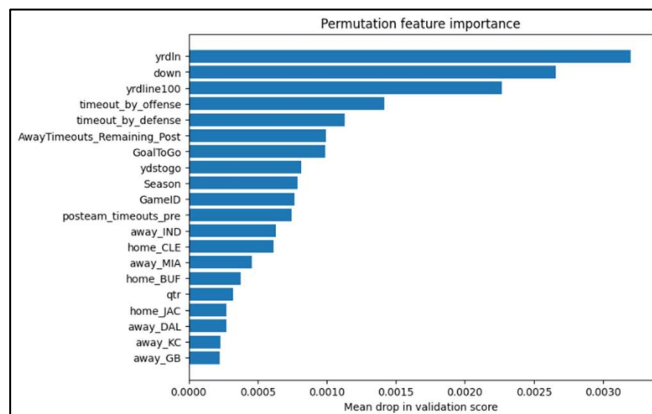


Figure 8: Model #3 Feature Importance

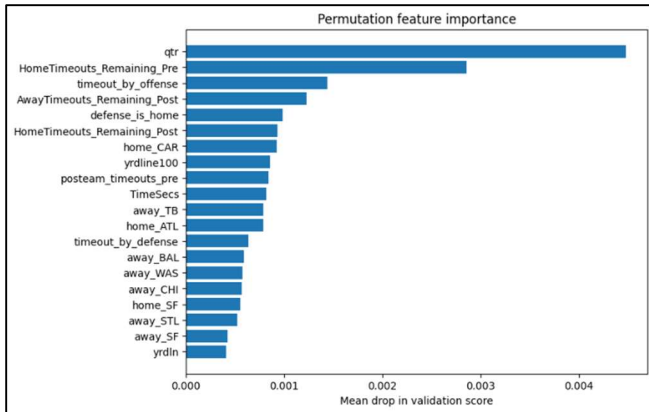


Figure 9: Model #4 Feature Importance

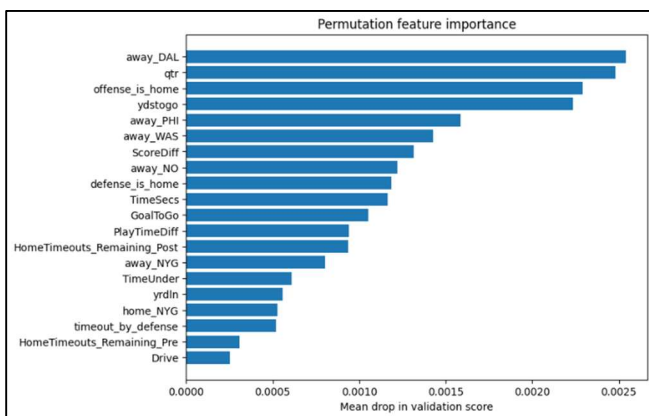


Figure 10: Model #5 Feature Importance

5. Discussion

The results from the models indicate higher error than I was hoping to get from the play-by-play data. This model performed between five and eight points worse per game and about 0.25 point per play than the best performing models in the wild that use aggregated game stats. However, the consistent results in loss and points across model runs gives me confidence that each of the first four models will generalize well to new data, even with questionable accuracy.

I would've liked to include team information more than home and away, but the dataset did not have stats aggregated by team. Additional pre-processing would have been required to get that information. The additional preprocessing could have been useful in other ways, as multiple

features that I expected to be important turned out to be important, but not necessarily in the ways that I expected. Some consolidation of those features could have ensured that they were included more purposefully and may have helped model performance; especially in the cases of timeouts used and position on the field.

6. Conclusion

The utilization of play-by-play data to predict expected points added, in addition to the inclusion of additional features that ultimately contributed little importance to the model, appears to have generated too much noise for the model to accurately predict expected points per play and game any better than existing models.

When considering that models #3 and #4 provided the greatest accuracy, the lower standard deviation within results from model #3 compared to model #4 indicates that my best performing model was model #3. However, all models, except for #5, performed similarly.

The features that seemed to be the most important in training consisted of 'down', yards to go, quarter, and some permutation of the yard line and timeout features, which I should have reduced to single features in preprocessing to reduce noise. This is expected because, as it was stated in the introduction, they were the game condition features that contributed the most to EPA calculation. It also shows that the model was able to pick out the right things to make its predictions.

Further research with this data set could include processing the data in such a way that important features could be combined to reduce noise in training the model. Adding features such as a value indicating the strength of each team and consolidating the features indicating the home and away team could also be helpful.

7. References

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- [2] A. Ennis, "An Introduction to Modeling NFL Tracking Data," *Yeshiva University*, 2020. [Online]. Available: <chrome-extension://efaidnbmnnnibpcajpcgclefindmka/j/https://repository.yu.edu/server/api/core/bitstreams/c4fb59c3-adc6-4e56-bf3f-b9618bd79fdf/content>. [Accessed: 31-Oct-2025].
- [3] J. Davis, L. Bransen, L. Devos, A. Jaspers, W. Meert, P. Robberechts, et al., "Methodology and evaluation in sports analytics: challenges, approaches, and lessons learned," *Machine Learning*, vol. 113, no. 9, pp. 6977-7710, Sep. 2024. doi: 10.1007/s10994-024-06585-0. [Online]. Available: <https://link.springer.com/article/10.1007/s10994-024-06585-0>
- [4] R. Yurko, S. Ventura, and M. Horowitz, "nflWAR: a reproducible method for offensive player evaluation in football," *J. Quant. Anal. Sports*, vol. 15, no. 3, pp. 163-183, 2019. doi: 10.1515/jqas-2018-0010. [Online]. Available: <https://www.degruyterbrill.com/document/doi/10.1515/jqas-2018-0010/html> [Accessed: 24-Nov-2025].
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8. Figures and Tables

Model #	Layers	Hidden Layers' Activation	Neurons Layer 1	Neurons Layer 2	Neurons Layer 3	Output Layer Activation	Output Layer Neurons
1	3	ReLu	16	8	N/A	Linear	1
2	4	ReLu	128	64	32	Linear	1
3	3	ReLu	4	6	N/A	Linear	1
4	3	TanH	4	6	N/A	Linear	1
5	3	TanH	4	6	N/A	Linear	1

Table 1: Model Configurations

Model #	Run 1 MAE/ Play	Run 1 MAE/ Game	Run 2 MAE/ Play	Run 2 MAE/ Game	Run 3 MAE/ Play	Run 3 MAE/ Game	Mean MAE/ Play	Mean MAE/ Game	Std Dev/ Play	Std Dev/ Game
1	0.9619	16.48	0.9599	16.70	0.9620	16.46	0.9613	16.55	0.001185	0.1332
2	0.9677	17.93	0.9587	16.14	0.9632	16.54	0.9632	16.87	0.004500	0.9395
3	0.9616	15.73	0.9519	15.95	0.9614	15.92	0.9583	15.89	0.005543	0.1193
4	0.9554	17.30	0.9688	16.27	0.9502	14.88	0.9581	16.15	0.009597	1.214
5	1.009	20.11	1.024	27.35	1.007	25.91	1.013	24.46	0.009292	3.833

Table 2: Model Results